

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Coded the document search engine for a debate research SaaS platform using *Python (FastAPI)* and *Elasticsearch*
- Collaborated with **10 engineers** in an *Agile* team on an *AWS*-hosted SaaS platform drawing **10K+ site visitors**
- Designed and built a text-retrieval algorithm that validates source URLs and filters statistical outliers with **IQR**
- Configured *Elasticsearch* to index **600K+** debate-evidence documents and return results in **under 1 second**

Product Development Intern | Jetson

January 2025 – February 2025

- Communicated recommendations directly to the CEO in a **5-page report** on strategies to optimize user engagement
- Applied UX design through **7 iterations of UI/UX testing** on **3 experimental features** for **100K+ users**
- Redesigned onboarding UX in *Figma*, cutting it from **12 to 5 pages**, and validated design with **A/B testing**

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed **200+ TikTok posts** spanning **3 years** with *Hootsuite*, *Microsoft Excel*, *Power BI* and *Python (pandas)*
- Found through data analysis that “science fun facts” posts earned **69.3%** of views from just **7%** of all content
- Persuaded the Executive Director to adopt a “fun facts” strategy that improved engagement **24%** (8K+ views)

PROJECTS

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Added a **RAG** agent (*Claude Sonnet 4.5*) that answers from session-scoped vector documents via *Supernemory*
- Integrated external *OpenAlex* and *Semantic Scholar* APIs into a data pipeline that backfills paper metadata
- Modeled paper data in *PostgreSQL* on *Supabase* and queried it with *SQL* by field, year and total citation count
- Developed a web UI on *React Flow* that graphs each paper’s outbound citations across **26 research fields**

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Built a truck-pricing MVP that uses a *JSON schema* to guide a *Claude Sonnet 4.6* agent in reading truck photos
- Applied prompt engineering to improve AI responses, refining the schema until **10 consecutive images** were right
- Audited data quality across **15,000** truck-listing images by de-duplicating listings and filtering non-USD prices
- Developed an analytical *quantile regression* model that estimates prices with **87% accuracy** on **30 samples**
- Won **2nd Place at UIUC Founder’s Hackathon** leading team of **4** on pricing algorithm for Kamion (YC S22)

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built a voice interface where a *Gemma Voice LLM* delivers steps from **4 Random Forest** anomaly models
- Directed a team of 5 from ideation and development to the final presentation. **USAII Global Hackathon Finalist**
- Wrote a *Python FastAPI* back end that ingests water, waste and energy data from all **3 tools** to detect anomalies
- Developed **3 edge AI tools** that collect data on a school’s energy, water and food use for anomaly detection

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Built a *React/Firebase* scheduling agent that books meetings for groups of **30+ people** on Google Calendar
- Designed a **5-step Python** scheduling algorithm that scans calendars, ranks time slots and emails confirmations
- Tested the algorithm across **14 unit and integration test cases** from availability scan to booking confirmation
- Signed **2 LOIs worth \$3.5K** and licensed Chronos to **8 startups**. **Won 2025 Congressional App Challenge**

SKILLS

Programming Languages: Python, SQL, JavaScript, Java, C++, HTML/CSS

Frameworks/Platforms: FastAPI, Next.js, React, AWS, Flask, Node.js, Supabase, PostgreSQL, Vercel, Firebase

AI/ML: Claude API, Hugging Face, Vercel AI SDK, Gemma 4, Supernemory, PyTorch, scikit-learn, TensorFlow

DevTools: Figma, Git, GitHub, Claude Code, Codex, Docker, CI/CD, VS Code, Bun, Clang, Catch2, Arduino IDE

Concepts/Practices: Generative AI, Prompt Engineering, Text2SQL, Data Engineering, LLMs, RAG